

Yingjing Huang

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Research Interest

GIScience, GeoAI, Deep Learning, Computer Vision, Urban Visual Intelligence, Social Sensing

Education

Peking University

Ph.D. in Geographic Information Science

• Advisors: Yu Liu, Fan Zhang

Beijing, China

Sep. 2021 - Jun. 2025

Wuhan University

MS. in Geographic Information Science

• Advisor: Teng Fei

Wuhan, China

Sep. 2018 - Jun. 2021

Wuhan University

BSc. in Geographic Science

• Advisors: Yiyun Chen, Teng Fei

Wuhan, China

Sep. 2013 - Jun. 2017

Experience

MIT Senseable City Lab

Visiting Ph.D. Student

• Advisors: Carlo Ratti, Fábio Duarte, Simone Mora

Cambridge, US

Jan. 2023 - Jan. 2024

Publications

Publications reported in reverse chronological order

(# co-first author, Google citations = 216)

Ongoing and Submitted Manuscripts

- [1] **Yingjing Huang**[#], Yong Li, Fan Zhang, & Yu Liu. Measuring sustainable development with urban imagery. (In preparing)
- [2] **Yingjing Huang**, Yuqing Wang, Ce Hou, Yanhua Yao, Fabio Duarte, & Fan Zhang. 2025. Revealing urban inequality from a dual perspective of indoor and outdoor spaces. *Landscape and Urban Planning*. (Under Review)
- [3] Yong Li, **Yingjing Huang**[#], Gengchen Mai, Fan Zhang. 2025. Learning street view representations with spatiotemporal contrast. *Computers, Environment and Urban Systems*. (Under Review, arxiv:2502.04638)

Peer-reviewed Journal Articles

- [1] Jing Xiao, Teng Fei, Bo Yu, **Yingjing Huang**, & Yunyan Du. 2025. Street view images help to reveal the impact of noisy environments on the survival duration of stroke patients. *International Journal of Health Geographics*, 24(1), 27. (IF = 3.2, Rank in Public Health, Environmental and Occupational Health = 106/687, Q1)
- [2] Yu Liu, Xuechen Wang, Yidan Wang, Fei Huang, **Yijing Huang**, Yong Li, Weiyu Zhang, Shuhui Gong, Gengchen Mai, Yao Yao, Yang Yue, Haifeng Li, & Fan Zhang. 2025. Representation learning for geospatial data. *Annals of GIS*, 1–27. (IF = 3.3, Rank in Earth and Planetary Sciences = 25/198, Q1)
- [3] Jin Zhu, **Yingjing Huang**, Ziyue Cao, Yue Zhang, Yuan Ding, & Jinglong Du. 2025. Evaluating urban greenery through the front-facing street view imagery: Insights from a Nanjing Case Study. *ISPRS International Journal of Geo-Information*, 14(8), 287. (IF = 2.8, Rank in Geography, Planning and Development = 95/841, Q2)

- [4] **Yingjing Huang**, Fan Zhang, Lun Wu, and Yu Liu. 2025. Measuring urban physical environments using image deep features. *Cities*, 166, 106196. (IF = 6.6, Rank in Urban Studies = 7/288, Q1)
- [5] **Yingjing Huang**, Fan Zhang, Yong Li, Lun Wu, & Yu Liu. 2025. Intelligent computational representation of urban imagery. *Geomatics and Information Science of Wuhan University*. (In Chinese, a leading journal in Geomatics in China, indexed by EI Compindex)
- [6] **Yingjing Huang**, Rohit Sanatani, Chang Liu, Yuhao Kang, Fan Zhang, Yu Liu, Fabio Duarte, and Carlo Ratti. (2025). No “true” greenery: Deciphering the bias of satellite and street view imagery in urban greenery measurement. *Building and Environment*, 269, 112395. (IF = 7.6, Rank in Geography, Planning and Development = 18/841, Q1)
- [7] **Yingjing Huang**, Fan Zhang, Yong Gao, Wei Tu, Fabio Duarte, Carlo Ratti, Diansheng Guo, and Yu Liu. 2023. Comprehensive urban space representation with varying numbers of street-level images. *Computers, Environment and Urban Systems*, 106, 102043. (IF = 8.3, Rank in Urban Studies = 3/288, Q1)
- [8] Meng Bian, Shuyi Guo, Wei Wang, Yuhui Ouyang, **Yingjing Huang**, and Teng Fei. 2021. Nextday forecast of Beijing pollen concentration fused with vegetation remote sensing data—Implementing a nonlinear autoregressive neural network model with external input. *Journal of Geo-Information Science*, 2021,23(09):1705-1713. (in Chinese, a key academic journal in GIS, indexed in the CSCD Core database)
- [9] **Yingjing Huang**, Teng Fei, Mei-Po Kwan, Yuhao Kang, Jun Li, Yizhuo Li, Xiang Li, and Meng Bian. 2020. GIS-based emotional computing: A review of quantitative approaches to measure the emotion layer of human–environment relationships. *ISPRS International Journal of Geo Information*, 9, 551. (IF = 2.8, Rank in Geography, Planning and Development = 95/841, Q2)
- [10] **Yingjing Huang**, Jun Li, Guofeng Wu and Teng Fei. 2020. Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus. *Cities*, 102, 102719. (IF = 6.6, Rank in Urban Studies = 7/288, Q1)
- [11] Yizhuo Li, Teng Fei, **Yingjing Huang**, Jun Li, Xiang Li, Fan Zhang, Yuhao Kang and Guofeng Wu. 2020. Emotional habitat: Mapping the global geographic distribution of human emotion with physical environmental factors using a species distribution model. *International Journal of Geographical Information Science*, 1-23. (IF = 5.1, Rank in Geography, Planning and Development = 31/841, Q1)
- [12] Shuangyin Zhang, Jun Li, Siying Wang, **Yingjing Huang**, Yizhuo Li, Yiyun Chen and Teng Fei. 2020. Repaid identification and prediction of cadmium–lead cross-stress of different stress levels in rice canopy based on visible and near-infrared spectroscopy. *Remote Sensing*, 12, 469. (IF = 4.1, Rank in General Earth and Planetary Sciences = 13/198, Q1)
- [13] Yiyun Chen, Tianci Qi, **Yingjing Huang**, Yuan Wan, Ruiying Zhao, Lin Qi, Chao Zhang, and Teng Fei. 2017. Optimization method of calibration dataset for VIS-NIR spectral inversion model of soil organic matter content. *Transactions of the Chinese Society of Agricultural Engineering*, 06, 107-114. (in Chinese, a top-tier journal in Agricultural Engineering in China, indexed by EI Compindex)

Patents

- [1] Fan Zhang, **Yingjing Huang**, Yu Liu. Method, Device, and Storage Medium for Comprehensive Landscape Representation Based on Street Imagery (in Chinese, National Invention Patent, China, Received, Not Yet Published)
- [2] Fan Zhang, **Yingjing Huang**, Diansheng Guo. Image Processing Method, Device, Product, Equipment, and Medium (in Chinese, National Invention Patent, China, Published, Not Yet Granted)
- [3] Fan Zhang, **Yingjing Huang**, Diansheng Guo. Method and Related Device for Identifying Regions in Space (in Chinese, National Invention Patent, China, Published, Not Yet Granted)

Honors & Awards

2022	China Scholarship Council (CSC) Scholarship for Visiting Doctoral Research A highly competitive national award to conduct one year of visiting research abroad	<i>Beijing, China</i>
2020	“Excellent graduate student” National award for the excellent graduate student in Wuhan University	<i>Wuhan, China</i>
2020	Kwang-Hua Scholarship Scholarship to the excellent student	<i>Wuhan, China</i>
2020	Second-Class Scholarship Awards to outstanding student in academic in Wuhan University	<i>Wuhan, China</i>
2015	Third-Class Scholarship Awards to outstanding student in academic in Wuhan University	<i>Wuhan, China</i>

Talks

Invited talks

Analysis methods and applications of street view imagery

Nanjing Normal University, China

[Link] (in Chinese)

Online

Aug. 2025

Intelligent computational representation of urban imagery

GISChat (A non-profit, GIS knowledge-sharing platform organized by Chinese scholars)

[Link] (in Chinese)

Online

Aug. 2025

Conference talks

Measuring urban physical environments using image deep features

The 26th Inter-University Symposium on Asian Megacities

Nanchang, China

Sep. 2025

Comprehensive urban space representation with varying numbers of street-level images

American Association of Geographers 2024 Annual Meeting

Honolulu, US

Apr. 2024

Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus

2021 International Graduate Workshop on GeoInformatics

Beijing, China (online)

Dec. 2021

Community Services

Journal Reviewer

- ISPRS Journal of Photogrammetry and Remote Sensing
- Landscape and Urban Planning
- Cities
- Computers, Environment and Urban Systems
- Urban Forestry & Urban Greening
- International Journal of Geographical Information Science
- Travel Behaviour and Society
- Applied Geography
- Scientific Reports
- IEEE Transactions on Human-Machine Systems
- Journal of Urban Technology
- Computational Urban Science
- Journal of Transportation Engineering and Information (in Chinese)

Grants

2025-2029	Key Program of the National Natural Science Foundation of China (No. 42430106) Spatio-temporal optimization with geographical artificial intelligence	<i>Contributor</i>
2024-2026	National Natural Science Foundation of China (No. 42371468) Representing Urban Physical Environment using Street View Imagery	<i>Contributor</i>
2022-2025	National Natural Science Foundation of China-Excellent Young Scientists Fund Program (Overseas) (No. 41901321) Geospatial Data Mining & Visual Intelligence	<i>Contributor</i>
2022-2023	International Research Center of Big Data for Sustainable Development Goals (CBAS) Key Technologies and Applications of Big Earth Data for Sustainable Development Goals	<i>Contributor</i>
2021-2022	Tencent Academic Research Fund (No. RAGR20210101) Identifying urban informal settlement using multi-source big geo-data	<i>Key Contributor</i>
2020-2022	National Natural Science Foundation of China (No. 41901321) Understanding the relationship between urban physical environment and urban mobility patterns quantitatively using street view imagery	<i>Contributor</i>
2017-2021	National Basic Research Program of China (No. 2017YFB0503602) Geospatial Big Data Mining and Spatiotemporal Pattern Discovery	<i>Contributor</i>

Skills

Programming	Python, Matlab, R, Javascript, LaTeX
Packages	Sklearn, Pytorch
Databases	MySQL, SQLite, MongoDB
GIS skills	ArcGIS, QGIS, Geopandas, Arcpy
OS	Windows, Linux, Mac OS
Languages	Mandarin Chinese (<i>Native</i>), English (<i>Advanced</i>)